Design of a crash box for automotive bumper system

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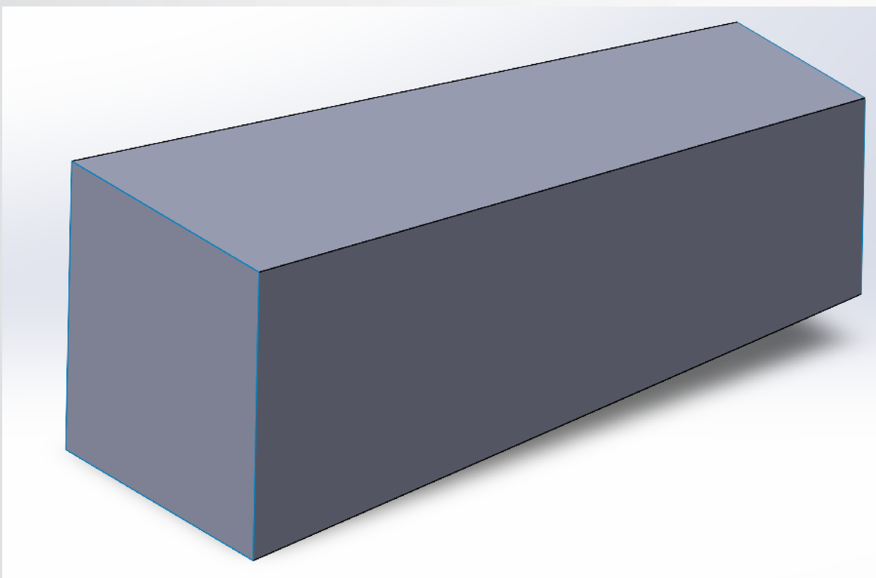
PROBLEM STATEMENT, CRITERIA AND CONSTRAINTS

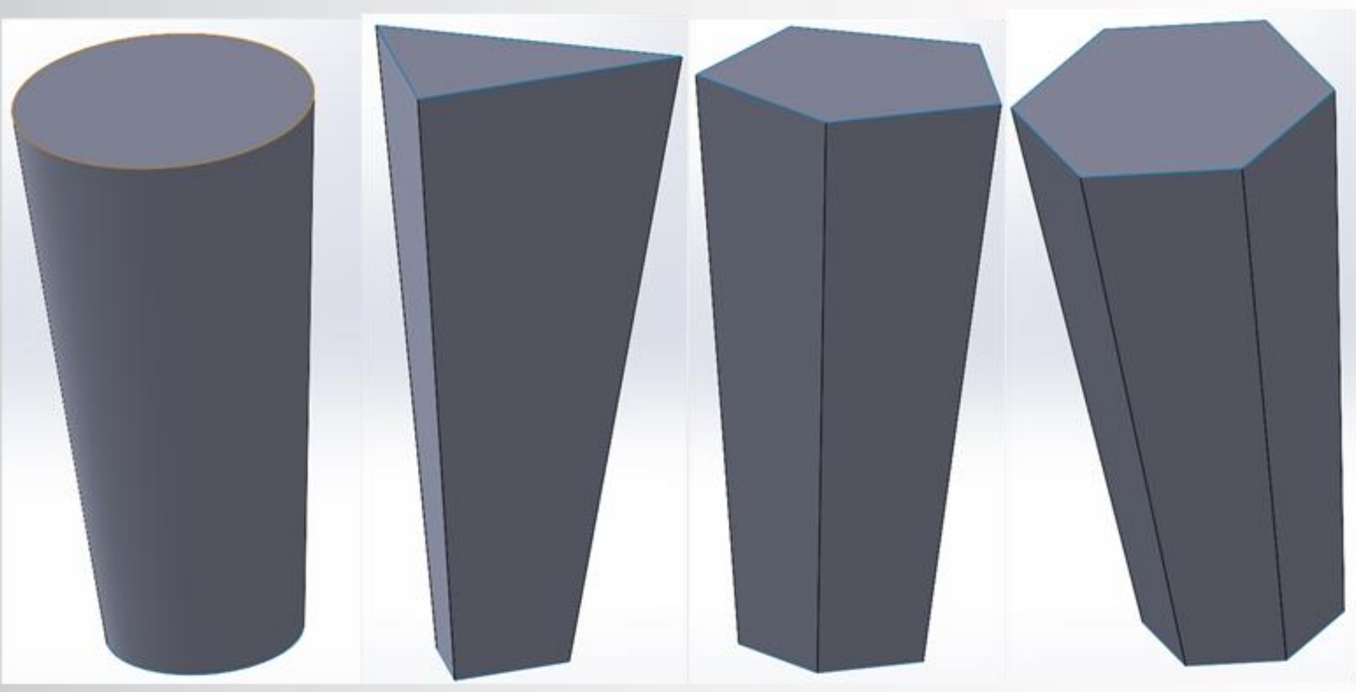
The imperative to enhance bumper systems for greater impact energy absorption and passenger safety is underscored by their current inefficiencies, which elevate risks of harm and repair expenses. The challenge lies in developing a crash box solution that enhances crashworthiness without inflating weight or costs. Criteria encompass heightened energy absorption, durability, and cost efficiency, all while upholding structural integrity and meeting stringent automotive safety standards. These objectives must be pursued within the constraints of minimizing weight and expenses while adhering to regulatory mandates throughout the design process.

EXISTING SOLUTION AND LIMITATION



Mamalis Model

PROPOSED SOLUTION & DESIGN



a) Circle b) Triangle c) Pentagon d) Hexagon

CONCLUSIONS AND FUTURE WORK

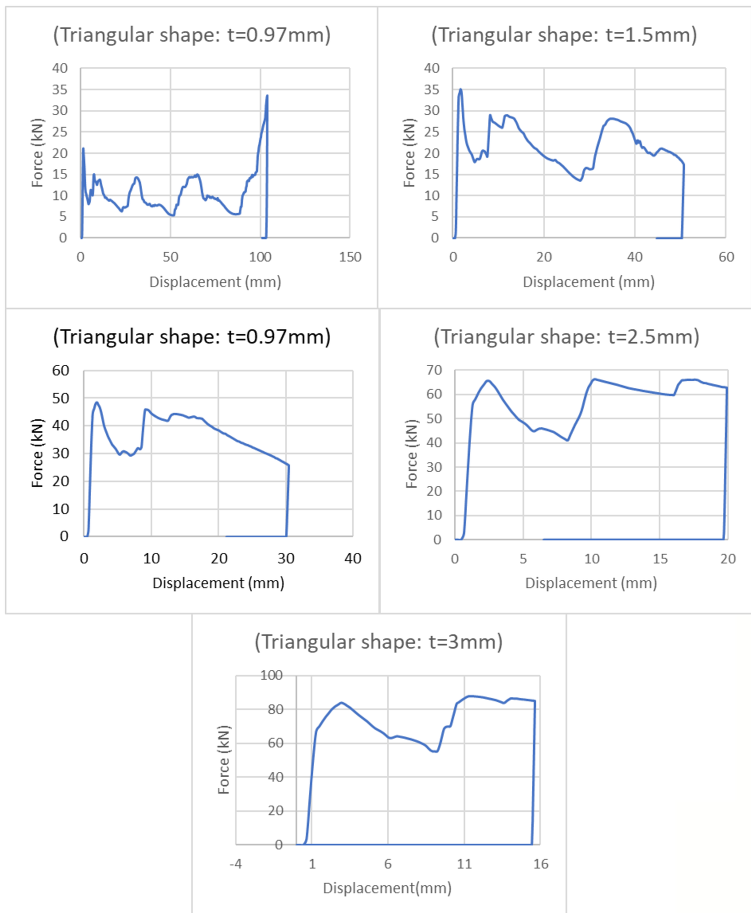
Crashbox configuration significantly impacts vehicle safety and structural integrity. The performance was refined through extensive research and modeling using SolidWorks, Hypermesh, and LS-Dyna. Optimal models, notably a triangle model with 2.5mm thickness, achieved the highest CFE value of 0.832, improving by 127% over the Mamalis model. A circular model, 3mm thick, exhibited the maximum performance with a SEA value of 6.853 kJ/kg. Future work should prioritize design optimization, physical validation, material exploration, and safety component integration.

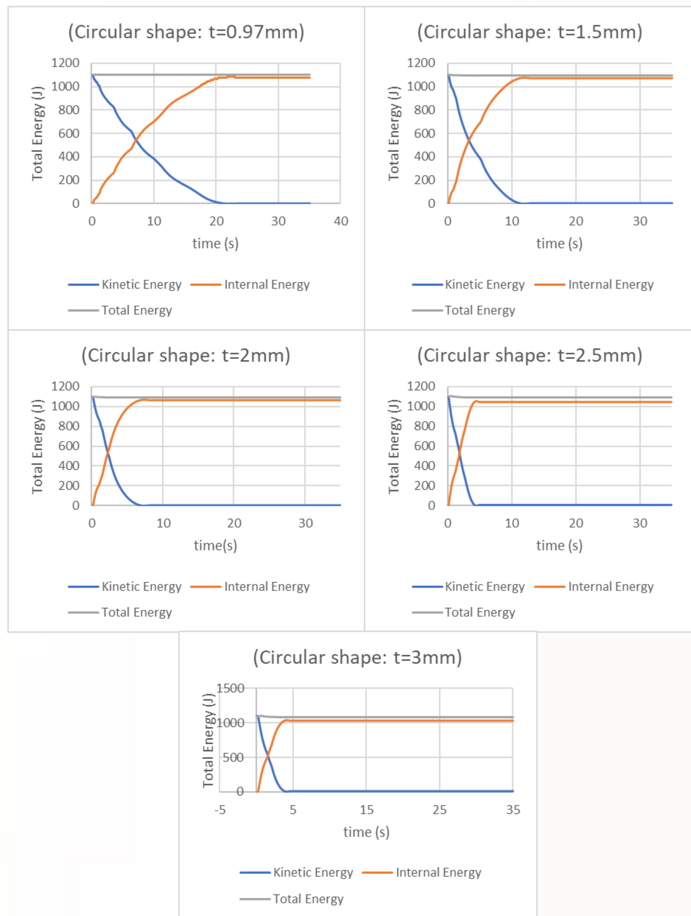
REFERENCES:

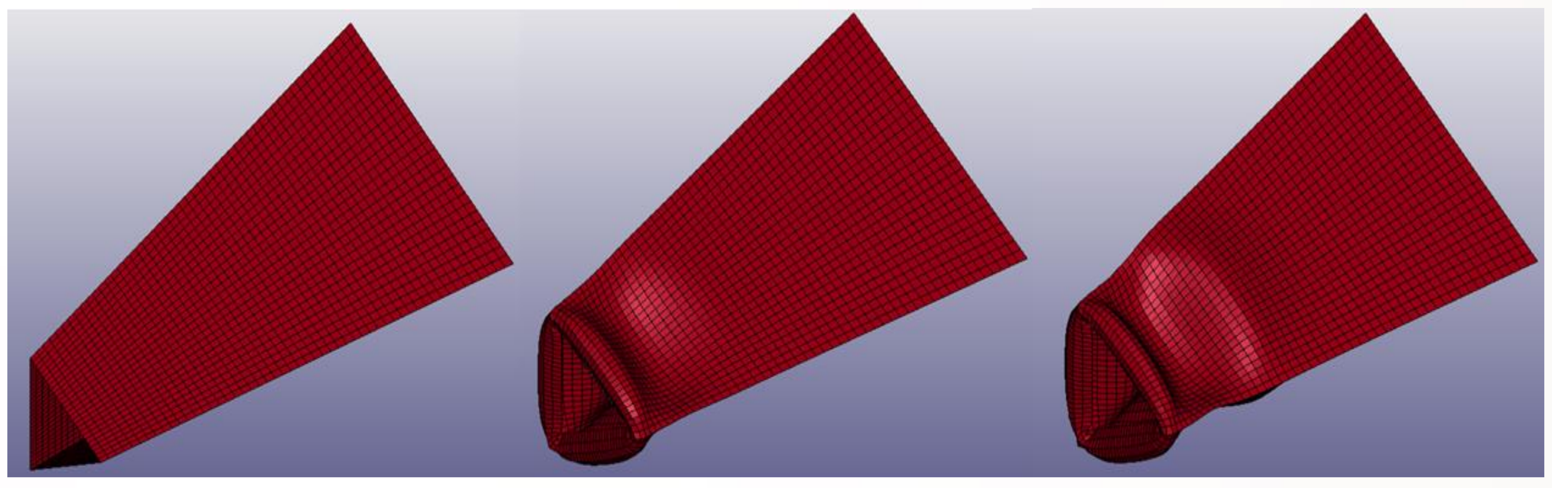
Guler, M. A., Cerit, M. E., Bayram, B., Gerçeker, B., & Karakaya, E. (2010). The effect of geometrical parameters on the energy absorption characteristics of thin-walled structures under axial impact loading. International Journal of Crashworthiness, 15(4), 377–390.

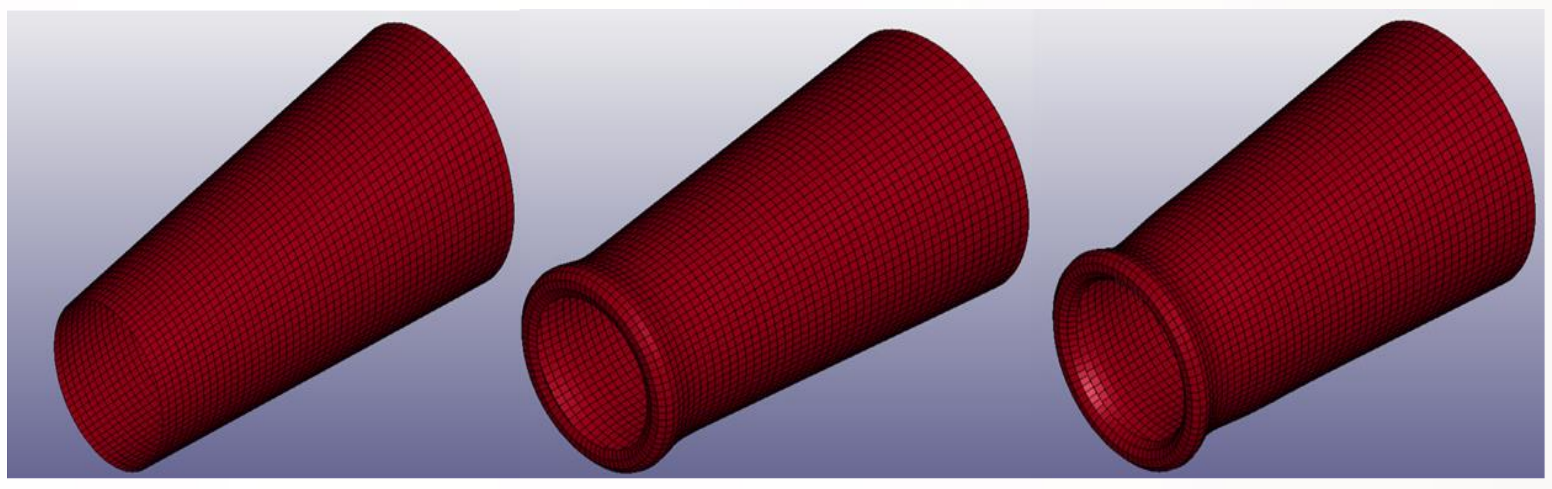
IMPLEMENTATION AND RESULTS

To understand how shape influences crashworthiness parameters, Solidworks models were created for different-shaped crashboxes with the same dimension of 0.97 mm. Next, the effects of additional thicknesses of 1.5 mm, 2 mm, 2.5 mm, and 3 mm were integrated into the analysis. Hypermesh generated the mesh, and LS-Dyna software completed the analysis with all required data and simulation runs, allowing a comparison between the different models.









The previous graphs show the effect of different thicknesses on the force vs. displacement relation for triangle shapes and total energy vs. time for circular shapes, as well as the deflection animation that occurred in the two best models according to CFE and SEA, which are the 2.5 mm-thick triangle and the 3 mm-thick circle, respectively.

Table 1: Crashworthiness parameters table for different models

Case	Mass (g)	max displacement (mm)	EA (kJ)	PCF (kN)	MCF (kN)	SEA (kJ/kg)	SEA % diff	CFE	CFE %diff
Mamalis	161.56	95.36	1.0988	31.44	11.52	6.801	-	0.367	-
0.97C	161.56	70.89	1.1000	29.30	15.52	6.809	0.11%	0.530	44.52%
1.5C	161.56	36.41	1.0995	50.25	30.20	6.806	0.07%	0.601	63.99%
2C	161.56	21.66	1.1012	71.74	50.84	6.816	0.22%	0.709	93.36%
2.5C	161.56	15.52	1.1036	104.39	71.11	6.831	0.44%	0.681	85.86%
3C	161.56	13.25	1.1071	113.10	83.53	6.853	0.76%	0.739	101.52%
0.97T	161.56	103.79	1.0983	33.60	10.58	6.798	-0.04%	0.315	-14.07%
1.5T	161.56	50.85	1.0986	35.04	21.61	6.800	-0.02%	0.617	68.25%
2T	161.56	30.50	1.0991	48.60	36.04	6.803	0.03%	0.742	102.34%
2.5T	161.56	19.94	1.0999	66.29	55.15	6.808	0.10%	0.832	127.02%
3T	161.56	15.64	1.1010	87.65	70.40	6.815	0.20%	0.803	119.16%
0.97P	161.56	77.42	1.0985	27.73	14.19	6.799	-0.03%	0.512	39.64%
1.5P	161.56	39.26	1.0989	45.19	27.99	6.802	0.01%	0.619	68.99%
2P	161.56	22.77	1.0993	61.78	48.29	6.804	0.05%	0.782	113.25%
2.5P	161.56	16.28	1.1002	87.45	67.56	6.810	0.13%	0.773	110.81%
3P	161.56	13.37	1.1016	110.78	82.42	6.819	0.26%	0.744	103.00%
0.97H	161.56	73.70	1.0984	28.55	14.90	6.799	-0.03%	0.522	42.46%
1.5H	161.56	35.79	1.0988	45.70	30.70	6.801	0.00%	0.672	83.31%
2H	161.56	22.27	1.0992	65.34	49.35	6.804	0.04%	0.755	106.09%
2.5H	161.56	15.84	1.1002	92.27	69.45	6.810	0.13%	0.753	105.37%
3H	161.56	13.10	1.1016	106.32	84.09	6.818	0.26%	0.791	115.79%

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